CHANDERNAGORE COLLEGE

SEMESTER – I B.A. / B.Sc. / B.COM (MAJOR/MINOR/SEC) INTERNAL ASSESSMENT EXAMINATION, 2023

SUBJECT :MATHEMATICS COURSE CODE : Math 1021 COURSE NAME : MINOR (4 & 3 YRS)

FULL MARKS: 10

TIME: 30 MINUTES

ATTEMPT ANY 02 FROM THE FOLLOWING QUESTIONS

2 x 5 = 10

Q1 :Show that the equation $(c^2 + d^2)(x^2 + y^2) = (cx + dy - cd)^2$ represents a parabola.

Q2 : If $I_n = \int_0^{\pi/2} x^n \sin x \, dx$, show that $I_n + n(n-1)I_{n-2} = n(\frac{\pi}{2})^{n-1}$.

Q3 :Prove that $\vec{a} \times (\vec{b} \times \vec{c}) + \vec{b} \times (\vec{c} \times \vec{a}) + \vec{c} \times (\vec{a} \times \vec{b}) = 0$.

Q4. If $y = x^{n-1} \log_e x$ then show that $y_n = \frac{n-1!}{x}$